

Fourier transform in QM

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By considering the extensive linear algebra framework of quantum mechanics, we can observe that the Fourier transform as a tool of quantum mechanics needs not to be explicitly defined, as it naturally exists within the framework itself. Firstly, consider a state vector $|\Psi\rangle$ expressed in the momentum basis:

$$|\Psi\rangle = \int dp \psi(p) |p\rangle \quad (1)$$

If this looks unusual, it is simply the spectral analogue to the discrete:

$$f(x) = \sum a_n \phi_n \quad (2)$$

Where $\{\phi_n\}$ is a complete basis.

Suppose we want to express $|\Psi\rangle$ in the position basis. To do this, we can project it onto $|x\rangle$:

$$\langle x|\Psi\rangle = \psi(x) = \int dp \psi(p) \langle x|p\rangle \quad (3)$$

But what is $\langle x|p\rangle$? We already know that $|p\rangle$ obeys the eigenvalue equation:

$$\hat{p}|p\rangle = p|p\rangle \quad (4)$$

Where $\hat{p}$ is given (in natural units) as:

$$\hat{p} = -i \frac{\partial}{\partial x} \quad (5)$$

We may project both sides of (4) onto $|x\rangle$:

$$-i \frac{\partial \langle x|p\rangle}{\partial x} = p \langle x|p\rangle \quad (6)$$

This differential equation is easily solved to be:

$$\langle x|p\rangle = A e^{ipx} \quad (7)$$

We set $A = 1$ here, as we only seek a functional form for $\psi(x)$. Substituting this into (3) yields:

$$\langle x|\Psi\rangle = \psi(x) = \int dp \psi(p) e^{ipx} \quad (8)$$

Which is exactly the Fourier transform of $\psi(p)$, implying that such transforms are fundamental to this mathematical framework.